On the Study of Different & Rotor Geometry Configuration for use in High-Speed Induction Motors

MASTER THESIS

In partial fulfillment of the requirements for the degree

“Master of Science in Engineering“

“Diplôme d’Ingénieur ESILV conférant le grade de Master“

Double Degree Master programs:

Mechatronik, Design & Innovation

MCI | The Entrepreneurial School®

Modeling & Computational Mechanics

Ecole Supérieure d’Ingénieurs Léonard de Vinci

Supervisor

Prof. Colin Stevens, Ph.D (MCI)

Prof. Colin Stevens, (ESILV)

Author

Colin Stevens

2314808 (MCI), 11111111111 (ESILV)

Preclusion from Public Access

I, Jason Brigham, have requested preclusion from public access for this master's thesis until 31.07.2031, which was approved by the study program.

Innsbruck, 14.5.2026

Signature

Declaration in Lieu of Oath

I, Jason Brigham, hereby declare, under oath, that this master's thesis with the title "On the Study of Different & Rotor Geometry Configuration for use in High-Speed Induction Motors" has been my independent work and has not been aided with any prohibited means. I declare, to the best of my knowledge and belief, that all passages taken from published and unpublished sources or documents have been reproduced, whether as original, slightly changed, or in thought, and have been mentioned as such at the corresponding places in the thesis by citation, where the extent of the original quotes is indicated.

I have only used the permitted and documented tools and have disclosed the use of generative AI systems, as required by the degree program. I am fully responsible for the selection, adoption, and all results of the AI-generated output I have used.

The paper has not been submitted for evaluation to another examination authority or has been published in this form or another.

Innsbruck, 14.5.2026

Signature

Abstract

Abstracts contain most of the following kinds of information in brief form. The body of your paper will, of course, develop and explain these ideas much more fully. As you will see in the samples below, the proportion of your abstract that you devote to each kind of informationand the sequence of that informationwill vary, depending on the nature and genre of the paper that you are summarizing in your abstract. And in some cases, some of this information is implied, rather than stated explicitly. The Publication Manual of the American Psychological Association, which is widely used in the social sciences, gives specific guidelines for what to include in the abstract for different kinds of papersfor empirical studies, literature reviews or meta-analyses, theoretical papers, methodological papers, and case studies.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 The Need for Speed

In recent years, improvements in manufacturing, transportation and process industry technologies bring about an increase in optimal operation speed in drive systems. In this respect, recently developed high speed gear-less or direct-drive electrical drives have seen an increase in interest based on the reduction in the total structural volume of the drive system. Due to the significant development of cost-effective, fast switching and compact variable frequency drives technology, wide speed range operations of different type AC motors has become feasible. The speed definition of an induction motor can be seen in Eq. 1.1,

$$n = \frac{120f}{p}. \tag{1.1}$$

In literature, there are several descriptions for the term "high-speed". As a mechanical engineer, peripheral speed over $150\,\mathrm{m\,s^{-1}}$ is considered to be high speed **gieras2011performance**. From the motor manufacturer's point of view, a two-pole machine which is supplied higher than 50 to $60\,\mathrm{Hz}$, can be considered as a high-speed machine. However, the most important point of view for the high-speed term is explained by development at power electronics. Nowadays, up to few hundreds hertz frequencies can be produced by variable frequency drives. However, voltage qualities of these are not satisfactory due to limited switching frequency of high-power IGBT technology. Thus, high-speed levels might be calculated for frequencies in the range of 100 to $400\,\mathrm{Hz}$

Table 1.1: As can be clearly seen, this table has absolutely no reason to be here aside from taking space. But it is a nice table to show how it should look and a template to write your own tables.

Admission requirements for Mathematics (MSc)	**Courses completed before start**	**Date of completion**
Mathematical Analysis (30 ECTS)	Mathematical Analysis 1	22.04.2014
	Mathematical Analysis 2	15.02.2013
	Complex Analysis	01.07.2015
Algebra/Linear algebra (22.5 ECTS)	Advanced Algebra	17.02.2013
	Abstract Algebra	01.06.2015
Geometry/Topology (15 ECTS)	Topology	01.11.2014
	Vector Analysis	15.06.2015
	Differential Geometry	15.02.2013

are considered to be high-frequencies **pyrhonen1991high**. Owing to brush and commutator structure causing mechanical and electrical problems, DC drives are not allowed to be used for high-speed applications. In addition to the aforementioned statement, the structure is not appropriate for large centrifugal forces. Nevertheless, as high-speed drive applications, there are different type of AC motor concepts proposed in literature **gieras2011performance**, **pyrhonen1991high**, **lahteenmaki2002design**, **saari1998thermal**: Laminated/solid induction, permanent magnet synchronous and switched reluctance synchronous motors.

It is due to remarkable improvements of power electronics, frequency inverters and AC variable-speed drives, which allows a wider use of applications of solid-rotor induction motor (SRIM). SRIM's are used in drive application ranging from a few kW's to 10 MW's. Fans, compressors, pumps, gas turbines, sewing machines, space and aeronautics, auxiliary motors for starting turbo-alternators, eddy current brakes, two-phase servomotors are a few examples of its areas.

Below are the advantages of SRIM compared to CRIM:

- Structural and Mechanical integrity, Rigiditiy, Reliability and Strength of Material
- High thermal properties
- High speed in high power applications (high moment density)

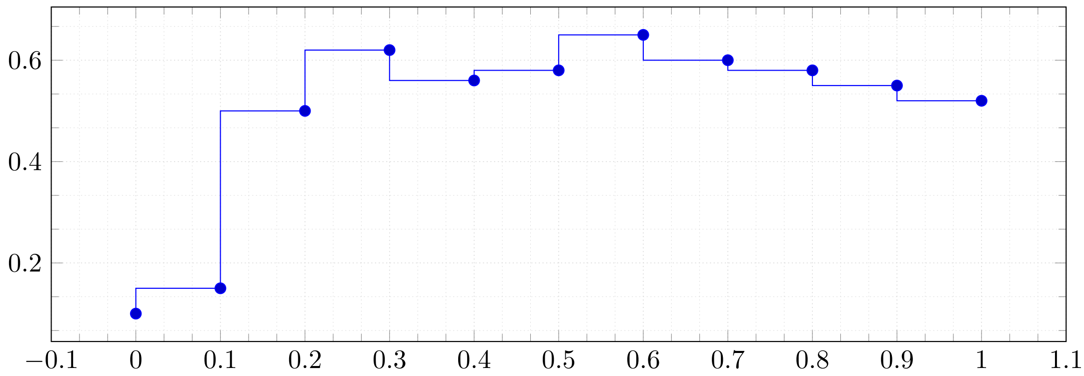

Figure 1.1: The authors interest in the topic as years go on.

- Low noise and vibrations in high speed applications
- Simple to protect against aggressive chemicals
- Ease of Manufacturing
- Low level of noise and vibrations (If the rotor has no slots)
- Linearity of torque-speed characteristics throughout the entire speed range
- The possibility of obtaining steady-state stability.

In 1950s, Solid-Rotor topologies for induction machines operating at high speeds have gained a lot of interest. From its inception to 1970s, various scientists and engineers have contributed to the development and the theory of solid rotor construction, where significant interest was seen the 1990s for using solid rotor structure for high speed applications.

As the rotor does not contain any trace of copper [1]windings, the eddy currents roam in the rotor without any conductive path restriction and cause the motor to have different characteristics compared to a Cage Rotor Induction motor (CRIM), an industry standard construction. While eddy currents are the main principle of its operation, these currents also causes the motor to have lower efficiency in slow speed aplications and this indirectly decreases its power factor. But in high speed application

[1]the rotor consists of a solid body of steel or similar ferromagnetic material

where the speed is around 30000 rpm the losses become far less and SRIM becomes the better choice for high speed applications.

1.1.1 Report Structure

In this report for Drive Technologies, the high-speed performance of the four different types of rotors are investigated and compared using finite element analysis: cage, smooth solid, axially slitted, coated is designed using a finite-element analysis (FEA). All rotors are designed with similar geometries and construction parameters to minimise the effect of unwanted effects.

Listing 1.1: Descriptive Caption Text

```
#include <bits/stdc++.h>
using namespace std;

// function find string greater than
// length k
void string_k(string s, int k)
{
    // create an empty string
    string w = "";
    // iterate the loop till every space
    for (int i = 0; i < s.size(); i++) {
        if (s[i] != ' ')

            // append this sub string in
            // string w
            w = w + s[i];
        else {

            // if length of current sub
            // string w is greater than
            // k then print
            if (w.size() > k)
                cout << w << " ";
            w = "";
        }
```

```
    }
}
```

The number of turns per stator slot is selected so as to obtain the same stator current in rated operation. All four motors were analyzed using FEA tools for 20 different speeds in order to obtain the combined torque-speed characteristics. To illustrate visually the differences in the distributions of the magnetic flux and the eddy current the results for 11300 rpm are presented and the core loss, the total loss and the efficiency for the specific speed are compared for all the motors. The designed four motors will be also compared for winding currents, induced voltages, flux linkages, electromagnetic torque, copper losses, iron losses, solid rotor losses and efficiency.

Chapter 2

Finite Element Method

2.1 Introduction

The determination of the solid-rotor induction motor performance has to be carried out through the solution of Maxwells field equations. In order to take into account the whole machine geometry, a large number of partial differential equations or integral equations have to be solved. An effective and widely used method of numerically computing eddy-current fields is the finite element analysis. For accurately solving distributed eddy-current problems containing partial differential equations or integral equations an effective computer program is a useful tool.

The comprehensive considerations and covering equations for the finite element method are reported for instance by [30], [31] and [32]. The leading process in FEM is subdividing the described machine geometry into portions for which a particular covering equation is valid. The described geometrical portions are further subdivided into a large amount of smaller elements, finite elements. The sizes of finite elements should be well proportioned in the motor geometry and, especially, the meshing of the air gap should be done carefully. The air gap is divided into three layers of equal width: the outer and inner layers are static and the middle layer is the dynamic layer in which the meshing is regenerated for each time step. In saturating parts such as a solid rotor the maximum element size is usually considered to be one third of the depth of penetration. Although a fine mesh-

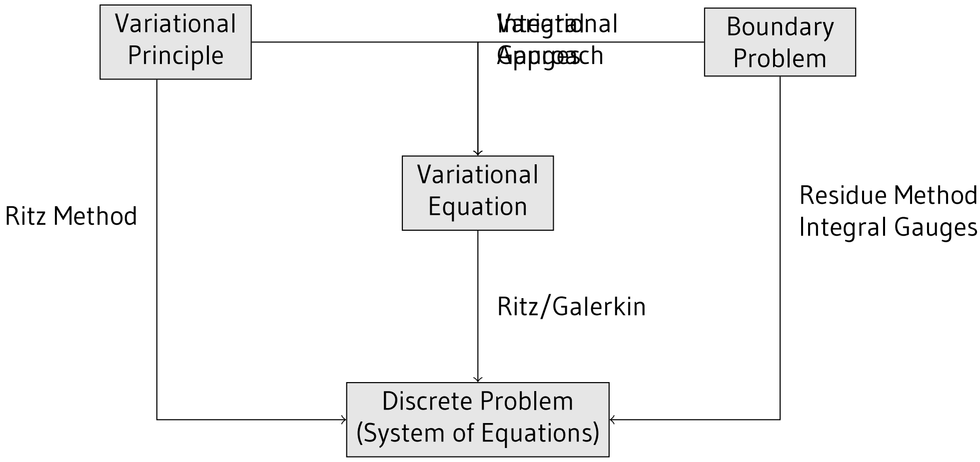

Figure 2.1: A Flow chart for finite element method.

ing requires a long computation time, special attention has to be paid also to the meshing of the rotor to accurately describe the complex eddy-current problem in the ferromagnetic solid rotor.

Therefore, a very dense meshing of the rotor and, in particular, on its outer layers is preferred.

2.2 Maxwell's Equations

The electromagnetic behavior exhibited during the normal operation of electrical machinery can be explained using Maxwells Equations. Faradays indution law determines the connection between the electric field strength E and the magnetic flux density B. Amperés law give that the relation between magnetic field strenth H and current density J and the changing of the electric flux density D.

$$\nabla \times E = -\frac{\partial B}{\partial t} \tag{2.1}$$

2.3 Modelling Losses

2.3.1 Stator Core Losses

Time-changing magnetic fields in ferromagnetic materials are the main reason for the magnetic iron losses happening in the motor. These losses include the hysteresis losses Phys which caused by the magnetic hysteresis loop of the ferromagnetic material and eddy current losses. These eddy current losses are separated into two types of losses which are as follows;

- Classical eddy current losses (Pclass)
- Excess losses (Pabnormal) which is mainly caused by non-uniform distributions of magnetic flux density in the lamination. This occurs because of the grain properties of the ferromagnetic material [16]

The hysteresis losses which are caused by the AC magnetization process are equal to the total area of the quasi-static hysteresis loop times the magnetizing frequency in addition to the volume of the core [18]. The total loss energy per cycle of the hysteresis loop can be seen in (3.4)

The hysteresis losses in a motor can be approximately calculated using Steinmetz formula which is based on empirical studies.

$$P_{hys} = k_h f B_m^2 \tag{2.2}$$

Where kh is the hysteresis constant which is related to the nature of the material used, f is the frequency and Bm is the maximum magnetic flux density. The eddy current losses that is occurred in the system is caused by the induced electric currents in the magnetic core by an external time changing magnetic field. These losses can be seen in (4.6) The eddy current loss constant, keddy, is related to the conductivity and the thickness of the material used [32]. Because of the difference of magnetic flux density in the yoke and teeth of the stator core, it is logical to divide the losses into two categories. Because of the way stators are constructed, some unwanted construction defects occur which in turn causes the increase in eddy current losses. In light of all this the total hysteresis and eddy current losses can be calculated using the equations given below; Where v10 is the total iron loss constant which is calculated in (4.9), Bteeth,1/3 and Byoke are the magnetic flux densities at the stator teeth at 1/3 of the tooth length and the stator yoke respectively , mteeth and myoke are the total mass of the teeth and the yoke respectively. bsh is the thickness of the selected loss stator laminated steel and kf is the lamination staking factor.

For use in the study, the stator iron losses are calculated using numerical methods using the Bertotti formula given in (4.10) [33]. So a non-sinusoidal time variation of B(t) can be calculated, including the effect of abnormal losses. In (4.10) it is considered, that B(t) is made up of a fundamental sinusoidal variation B(t) with the frequency f and the deviation of that does not affect the hysteresis losses. Where p(t) is the specific iron loss per volume of the iron core, kf is the lamination stacking factor, kh is the hysteresis loss constant, Bm is the maximum flux density of the fundamental sinusoidal time variation, f is the operation frequency, is the electrical conductivity of the core sheets, bsh is the lamination thickness and kex is the abnormal eddy current losses constant.

2.3.2 Rotor Eddy-Current Losses

When the systems are modelled in FEM, the time variation of the stator voltage and the current are given in pure sinusoidal forms. When the spatial distribution of flux density that occurs in the air gap is concerned a pure sinusoidal hypothesis would not work. The space distribution of the flux density has no correlation to the time variation of the time current that is being fed to the windings. This phenomena occurs only on the winding arrangement in the slots and on the rotors own geometry.

The rotor eddy current losses Peddy is made up from the fundamental component P., which dependent of the rotor frequency, and high frequency eddy current losses Pftr,add . Which is caused by slotting [34]. These rotor eddy current losses were numerically calculated with the aid of FEM simulation using (4.11) [20, 21]. In (4.11) the constant represents the specific electrical resistance of the rotor steel. The calculations were done with the aid of FEM. The total amount of eddy current loss density that occurs in the rotor can be categorized in three sections;

- High value eddy current loss beneath the rotor surface with a penetration depth of d

- Eddy current loss that occurs in the teeth
- Eddy current loss that occurs in the yoke

In light of all this the total loss of eddy currents can be summed in (4.13)

2.3.3 Joule (Electrical) Losses

Electrical or Joule losses happens when the current passes through a conductor and because of the conductors resistance gives of energy in term of heat. These losses include the current with the fundamental and harmonic frequencies and also include the losses that is caused by the current displacement. The total amount of winding losses are the sum of I2R (ohmic) losses , the 1st order current displacement losses Pcu,1 which happens because of the unbalanced current sharing between the parallel wires per turn and finally the 2nd order displacement losses which occurs because of the skin effect within each wire.

Rs, which is the phase resistance, is in direct relation with the operating temperature. When the motor is in operation, the current I flows through the winding conductors. These current that flow through the conductor, depending on the frequency and the geometry, induce eddy currents mainly in the slot part of the conductors. Due to these eddy currents, the total conductor current density J is displaced within the conductor cross-section. This phenomena is also known as skin effect. In relation to all aforementioned above, the second order current displacement causes the increase on Stator AC resistance which in turn increase the Joule losses. In addition, the first order current displacement losses happens because of the uneven distribution on parallel wires per turn which can me measures given in [35][36];

$$R_S = R_{S,\,20}\left(1 + \frac{\theta - \theta_0}{235 + \theta_0}\right) \tag{2.3}$$

In the equation given above, RS,20 is the phase winding dc-resistance at V0 = 20 and is the winding operating temperature While high speed motors can operate at higher frequencies than conventional motors, the current displacement that is caused by the internal eddy currents are more prominent at loss calculations. When a Dc current is applied it effect the conductor evenly but, when an AC current is applied this is not the case (Skin Effect). The current penetration depth is defined usually as the main boundary of the region, the part where the most of the current flows. Since this area is only some part of the conductors area, the resistance for the AC current is higher which also effects the first and the second order current displacement. This increase in AC winding resistance also directly affect the winding losses, which are calculated in (4.16) and (4.22) [37]

Chapter 3

Conclusion and Future Work

In this study, performances of four different types of rotors which are cage, smooth solid, slitted solid and coated solid rotor were investigated for high speed drive applications. Firstly, basic information about four different rotor types was represented. All 2D transient magnetic models of high speed induction motors were built up in FEA tool. All motors having different rotor types have the same stator core with different numbers of turns per slot. The other geometrical properties were remained constant. All motors were analysed for numerous speed levels and integrated torque-speed characteristics were obtained and compared.

Distribution of magnetic flux density, magnetic flux lines and distributions of eddy current density of all rotor types were presented. Core loss, total loss and efficiency comparison of different type of rotors are given.

Although CRIM has a higher efficiency and the least loss, its mechanical output is not as high as other analysed motors. All SRIM provides more output power and torque for high speed applications. Inside the SRIM types CCSRIM serves the best torque-speed output for high-speed applications with the lowest core- loss and total loss and the highest efficiency beside its well-known physical and thermal capabilities.